# Tasnim Amara

Tasnim Amara | Portfolio | tamara@uwaterloo.ca

## Skills

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|---|---|
| Programming | Python, Java, MATLAB, R, HTML/CSS, Three.js |
| Software | Microsoft Office, KiCAD, SolidWorks, COMSOL, SIMULINK, Isaac Sim, Blender |
| Technical Skills | PCB design, Soldering, MCUs, Sensor Integration, Multimeter/DMM, Oscilloscope, 3D Modeling |
| Lab Techniques | Characterization, Organic Synthesis, Simple Distillation, Recrystallization, LLE, Vacuum Filtration, TLC, Micropipetting |

## Education

**B.A.Sc Honours Nanotechnology Engineering** **2025-2030**
University of Waterloo

## Experience

**Electrical Engineer** **Sept 2026 - Present**
*Waterloo Rocketry*

- Engineering a TDLAS (Tunable Diode Laser Absorption Spectrometer) payload for the team.
- Developing the schematic and PCB layout in KiCad for a main-board built around an STM32H503CBT6 MCU, which will power a laser and two photosensing boards, collect their data, and log it to an SD card.
- Designing SPI and CAN interfaces and inter-board connectors to relay data between the boards and the rest of the rocket.

**Materials Lab Volunteer** **Sept 2026 - Present**
*Multi-Scale Additive Manufacturing Laboratory*

- Co-audit 1,700+ container chemical inventory for compliance with life cycle documentation standards.
- Review and mark up existing standard operating procedures (SOPs) and risk assessments for internal consistency and compliance with Nanotechnology and Etching Lab safety requirements.
- Perform preventive coolant changeovers on precision saw, cut-off saw, and bandsaw equipment to support safe machine operation.

**Research Assistant Co-op** **May 2026 - Aug 2026**
*Waterloo Institute for Nanotechnology*

- Conducted literature reviews on spectroscopic (Raman, hyperspectral) and microwave sensing techniques for microplastic detection in water bodies.
- Evaluated detection methods for differentiating microplastics from sediment/silt based on feasibility and technical constraints.
- Presented findings and technology comparisons to lab supervisors to guide detection method selection for the collection device the team was building.

**Summer Reading Club Facilitator** **Jun 2025 - Aug 2025**
*Waterloo Public Library*

- Acted as the welcoming face for families, managing registrations and inquiries, directly increasing program retention through personalized interactions.
- Collaborated with a team to plan and execute a diverse schedule of 4 weekly activities, demonstrating strong project coordination and time-management skills.
- Coordinated a large-scale end-of-summer party for over 200 attendees, improving logistical planning and teamwork abilities.

# Projects

**Firefighter Flame Extinguisher Robot** **Microcontrollers, PCB Design, Troubleshooting**

- Built an autonomous fire-extinguishing robot that detects flames using IR/ultrasonic sensors and navigates to the source for 100% consistent flame detection using PIC16f18875 microcontroller.
- Designed custom PCBs from schematic using Traxmaker PRO and systematically troubleshot circuits with multimeter to ensure full functionality.
- Programmed MCU with autonomous logic in PICBasic Pro to identify flames and activate suppression. Developed pathfinding routines using ultrasonic/IR sensors for 100% consistent flame detection.

**Triboelectric Nanogenerator (TENG)** **SolidWorks, 3D Modeling, Prototyping**

- Designed and modeled a vortex-whistle TENG in SolidWorks to convert wind energy into electricity for portable lighting.
- Prototyped via 3D printing, optimizing materials for triboelectric output (0.4V measured).
- Collaborated in a team to iterate design and prepare technical documentation.

**High School Garden** **Project management, Collaboration**

- Awarded $1,000 grant through the Youth Creativity Fund to design and build a school garden aimed at improving school aesthetics in my senior year.
- Collaborated with a partner to manage the entire project budget, researching and sourcing materials to maximize value while staying under budget.
- Led and supervised a team of student volunteers during the construction phase, delegating tasks and coordinating logistics to ensure efficient workflow.